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Regeneration of Free Reducing Glycans from Reductive Amination-Tagged Glycans by Oxone

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Abstract

Glycans are usually fluorescently tagged by reductive amination for analytic tools. However, free reducing glycan regeneration is sometimes important and necessary for further structural or functional studies. Here, we introduce a new method for efficiently removing fluorescent tags from glycoconjugates by a simple treatment with Oxone. This method is proven to be fast and general after being tested on a series of common saccharides and widely used tags. We successfully achieved *N*-glycopeptide synthesis by using the regenerated glycans.

Graphical Abstract

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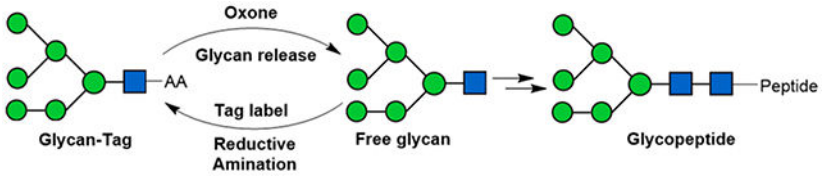
Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c02709.
Experimental section, copies of mass spectra, and copies of ^{1}H and ^{13}C NMR spectra (PDF)

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.1c02709

The authors declare the following competing financial interest(s): X.S. is co-founder of NatGlycan LLC, which is commercializing the ORNG technology.

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Functional glycomics has been attracting significant interest from the scientific community. The glycan moieties of glycoproteins, proteoglycans, and glycolipids on the cell membrane have proven to be functionally important through their interactions with other biomolecules such as glycan-binding proteins (GBPs).[1–9] Due to the non-template-driven biosynthesis, glycans are inherently complex and diverse, which has greatly hindered studies of their structure and function. Additionally, compared with those of linear biomolecules such as proteins and nucleic acids, the sequencing and preparation of glycans lag far behind. Because of recently developed analytical methods and technologies, such as high-resolution and specialized mass spectrometry (MS), high-performance liquid chromatography (HPLC), genetic and cell biological approaches, and glycan microarrays, it has become possible to explore glycan structures and functions more deeply.[10,11] However, one crucial barrier to the field is inadequate access to biomedically relevant and diverse complex glycan structures, which not only limits the development of analytical methods relying on structural standards but also directly limits the functional study of individual glycan structures. The synthesis of glycans has significantly improved along with the recent reports of the elegant synthesis of complex glycans and microwave/HPLC-assisted machine-driven glycan synthesis.[12–21] However, the synthesis of complex glycans is still challenging and laborious, which is practiced in only a limited number of highly specialized laboratories.

Isolation of diverse and biologically relevant complex glycans from natural sources is an alternative and feasible route.[18] Free reducing glycans, released and enriched from natural sources, are usually derivatized for commonly used analytic tools such as MS, HPLC, and microarrays.[19,20] Reductive amination is one of the most widely used methods for installing a hydrophobic and fluorescent tag at the reducing end of the free glycan. Typical reductive amination reagents include 2-aminopyridine, 2-aminobenzoic acid, aminoacridone, etc.[22–30] Under certain situations, free reducing glycan regeneration is necessary for further structural or functional studies after the tags have fulfilled their mission. On one hand, the derivatized tag often complicates permethylation of glycans for detailed structural analysis, although some of the tags can increase the MS sensitivity of free glycans. On the other hand, the tag itself, especially aromatic tags, might interfere with binding to GBPs during functional studies, such as microarray or cellular uptake experiments. Moreover, it is difficult to compare glycans with different tags in functional analysis. An efficient and practical tag removal method is thus in high demand. Even though several cleavable linkages have been developed in recent years,[31,32] many of these are based on hydrazone or oxime formation. These derivatizations are complicated by the formation of open-ring and closed-ring products (in α and β configurations), which are not ideal for the separation of individual glycans from a complex mixture. Reductive amination degenerates these anomeric isomers and thus facilitates glycomic analysis. However, few nongeneral methods were reported for reductive amination tag cleavage, all of which are inefficient.[33,34]

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We have recently reported a chemical approach for regenerating free reducing glycans from reductively aminated glycans by *N*-bromosuccinimide (NBS).[35] While this reaction is simple and fast, the major product of the NBS method is glycan with a pentose reducing end (Figure 1, ~1:3 hexose:pentose ratio for the AEAB tag), which is useful for only MS analysis. Furthermore, the pentose reducing end glycan is hard to separate from the desired intact reducing glycan through either a C18 solid-phase extraction (SPE) column or HPLC, because of their structural similarity and the lack of chromophores for detection. To address these issues, here we report the development of a facile and more efficient chemical method for removing fluorescent tags installed by reductive amination to generate intact free reducing glycan.

Release of Free Reducing End Glycans from Glycan-AEAB via Treatment with Oxone.

Inspired by the oxidative cleavage by NBS treatment, we explored other oxidative reagents and surprisingly discovered that Oxone (trade name $2KHSO_5{\cdot}KHSO_4{\cdot}K_2SO_4$) is potentially a better oxidative reagent for efficiently releasing glycans from reductively aminated glycan. Apart from oxidizing the amine group to the nitro group, it was reported that Oxone does not affect other common functional groups on glycan such as hydroxyl, acetyl, and sulfate groups.[36]

First, we treated AEAB-labeled glucose with Oxone as a model reaction, as shown in Figure 2a. When 10.0 mg of Glc-AEAB was treated with 10 mL of 1% Oxone in a 0.1% TFA solution for 0.5 h, the starting material disappeared nearly completely as monitored by NMR (Figure 2b). The whole reaction mixture was purified through a reverse-phase C18 SPE cartridge. We were able to recover free glycans (glucose and arabinose) from the flow-through fraction (H_2O) and the pure side product from the eluted fraction (25% ethanol in H_2O). Although arabinose is still observed as a side product, the ratio of the pentose reducing end (28%) is much lower than that from NBS treatment. The other side product is characterized as Glc-isoxazolidinone, which can be easily separated from free reducing glycans due to the higher hydrophobicity of the modified tag.

Release of Free Reducing Glycans from Fluorescent Glycoconjugates Prepared by Reductive Amination.

To explore the general applicability of this approach, we tested a panel of glycans, including glucose, GlcNAc, GalNAc, and lactose, and several common reductive amination tags. The selection of these mono- and disaccharides is modeled toward *N*-glycans (the reducing end is GlcNAc), *O*-glycans (the reducing end is GalNAc), human milk glycans, and glycans from glycosphingolipids (the reducing end is lactose or glucose). The results showed that Oxone-catalyzed free reducing glycan regeneration can be applied to all tested glycans and most reductive amination tags (Table 1) except the 2-PA tag. Glycan-PA will not react with Oxone at room temperature, probably because the pyridine structure will convert to *N*-oxide pyridine and prevent further reaction. In particular, only desired free GlcNAc/GalNAc and the corresponding side product glycan-isoxazolidinone were generated from the

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reaction without any significant amount of the pentose reducing end byproduct, regardless of different tags. Here the neighboring NH-acetyl group suppressed the generation of the pentose reducing end. As a result, the purification of regenerated *N*-glycan and *O*-glycan from reaction mixtures can be easily achieved with a simple C18 SPE column.

The results also showed that AA-tagged glycans generated higher yields of free hexose reducing end glycans than AEAB-tagged glycans with a >60% yield for all tested glycan-AA. Surprisingly, the glycan-AA/AEAB/MAA/AB generated the same side product glycan-isoxazolidinone regardless of the substitution on the carboxylic acid. The mechanism of generation of the side product glycan-isoxazolidinone was proposed on the basis of this observation. Additionally, by using an aniline tag, the byproduct glycan-isoxazolidinone could be completely avoided presumably due to no carboxylic group or carboxylic acid derivative being present at the *ortho* position of the amino group to form isoxazolidinone, and we even achieved a 100% yield of free reducing glycan for aniline-tagged GlcNAc and GalNAc. Thus, *m*- or *p*-aminobenzoic acid may be a better choice for future reductive amination fluorescent tag design.

One major purpose of removing these tags is to prepare complex natural glycans with a free reducing end. Therefore, we further tested this approach on complicated oligosaccharide conjugates to prove its general applicability (Figure S1). LC-QTOF-MS (liquid chromatography quadrupole time-of-flight mass spectrometry) analysis of the product of $Man_5GlcNAc$-AA with Oxone after C18 SPE showed a major product at $[M + Na]^+$ and $[M + K]^+$, which matches with the mass of free reducing $Man_5GlcNAc$. The Oxone treatment product was relabeled with AA, and the mass spectrum of the major product matches that of the starting material. The retention time of the AA-relabeled product on C18 HPLC also matches that of the starting material $Man_5GlcNAc$-AA. We have also examined the Oxone glycan release method on glycan containing acid-labile sialic acid (Figure S2), which proves that the Oxone release method is compatible with sialylated glycans.

Mechanism of Oxone-Catalyzed Tag Removal.

On the basis of three major products and former reports of a secondary amine treated with a peroxide reagent,[37,38] this simple and efficient reaction is likely to go through a mechanism of oxidation followed by elimination (Figure 3). In the first step, the C1–N linkage is oxidized to nitrone. Then β-elimination of H_2O is followed by imine hydrolysis to yield intact free reducing glycan. A six-membered-ring pericyclic rearrangement eliminates H_2O while breaking the C1–C2 bond, generating the pentose reducing end. This side reaction apparently is greatly suppressed by the NH-acetyl group compared to the hydroxyl group at C2. This is important as most natural glycans such as *N*- and *O*-glycans are terminated with HexNAc at the reducing end, making this method especially appealing in regenerating intact *N*- and *O*-glycans. The glycan-isoxazolidinone side product is generated by an internal five-membered-ring formation, which yields a product stable toward further oxidation and degradation. The lack of this side product when aniline-tagged glycans are treated with Oxone is consistent with the proposed mechanism. Interestingly, this side product has a stable closed-ring form similar to natural glycoconjugates, which could be useful in certain

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functional analyses that are sensitive to reducing end-ring opening. Because the proposed mechanism is not directly related to the structures of the glycans other than the reducing end and is general toward most of the tags tested above, we expect that this method should be a general method for most glycan conjugates prepared by reductive amination.

Application of Regenerated Free Reducing End Glycans.

Using this new method, isolated glycans tagged by reductive amination for HPLC separation should be easily converted back to free reducing glycans for further analysis and application. This strategy is especially suitable for tag removal of *N*- and *O*-glycans terminated with GlcNAc and GalNAc, which generate no significant amount of the pentose reducing end under treatment with Oxone. The only significant side product, glycan-isoxazolidinone, can be easily separated by C18 SPE. In our recently developed oxidative release of natural glycans (ORNG), one major side product consists of *N*-glycans missing the reducing end terminal GlcNAc. These glycans, after tagging and HPLC separation, can be used to recover reducing mono-GlcNAc *N*-glycans, which are important substrates for *N*-glycopeptide or glycoprotein assembly catalyzed by a series of endoglycosidase mutants (glycosynthases).[39–41] We have recently prepared significant amounts of high-mannose *N*-glycans (≤125 mg) from soybean protein isolates.[42] We treated purified $Man_6GlcNAc$-AA with Oxone to regenerate free $Man_6GlcNAc$ (60% yield). The regenerated $Man_6GlcNAc$ was then converted into the oxazoline form and conjugated with a GlcNAc-peptide, which is synthesized by Fmoc solid-phase peptide synthesis (SPPS) on an automated synthesizer, to form an intact *N*-glycopeptide. The conjugation reaction was catalyzed by an *Arthrobacter protophormiae* endo-*β*-*N*-acetylglucosaminidase mutant (EndoA-N171A).[43]

The final conjugated product is a homogeneous glycopeptide with defined glycan and peptide sequences, which was confirmed by MALDI-TOF-MS (Figure 4). This strategy can be potentially applied for the synthesis of homogeneous *N*-glycoproteins.

In conclusion, we developed a new method for efficiently removing fluorescent tags from glycans installed by reductive amination through a simple treatment with Oxone to recover free reducing glycans. This method is general to most tested tags, including all 2-aminobenzoic tags and the aniline tag. The regenerated free reducing glycans can be chemically modified at the reducing end for different purposes, including serving as a substrate for *N*-glycoprotein and *N*-glycopeptide synthesis. In our effort to isolate natural glycans for functional study, we routinely release and tag glycans for HPLC purification of individual glycans. Therefore, the strategy combining glycan release, tagging, and HPLC separation followed by Oxone treatment could serve as a general approach for the harvesting of free reducing glycans from natural materials. The Oxone reaction presented here is fast, general, and more efficient, and the purification of free reducing glycans is straightforward. It essentially renders the commonly used reductive amination a reversible process. We believe this method will greatly facilitate the isolation of free reducing complex glycans, which are highly desired resources for glycoscience study.

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Supplementary Material

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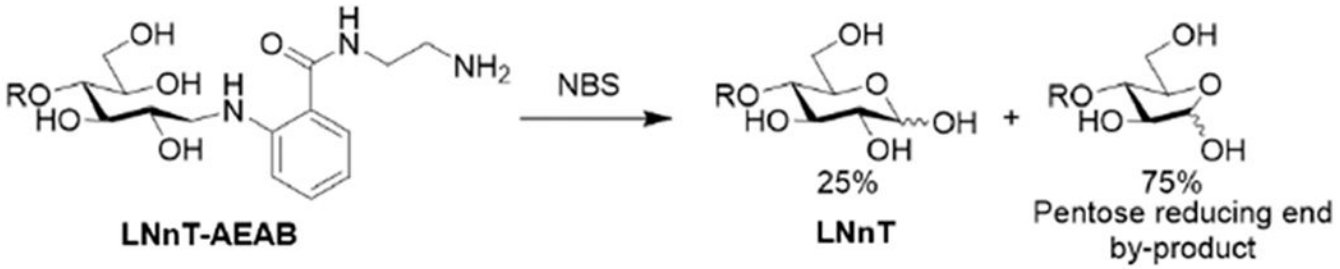


Figure 1.
Previous regeneration of free reducing glycan from a glycan tag by NBS.

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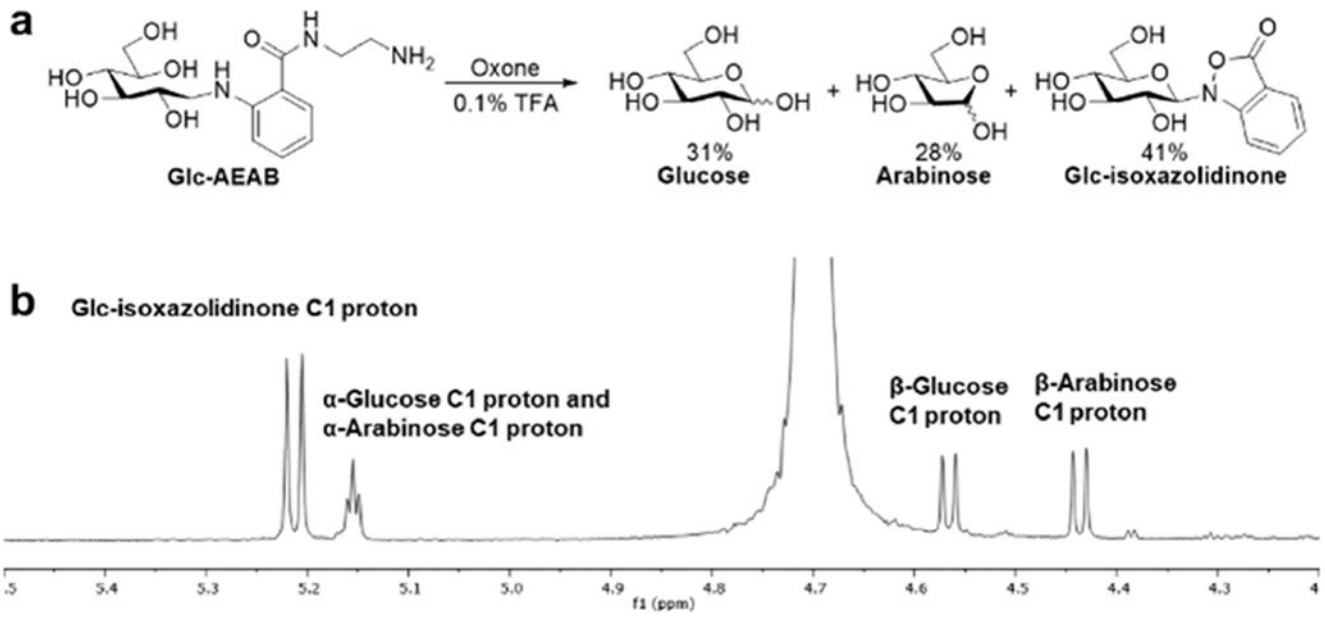


Figure 2.

(a) Reaction scheme and (b) ^{1}H NMR spectrum of Glc-AEAB treated with 1% Oxone in a 0.1% TFA solution.

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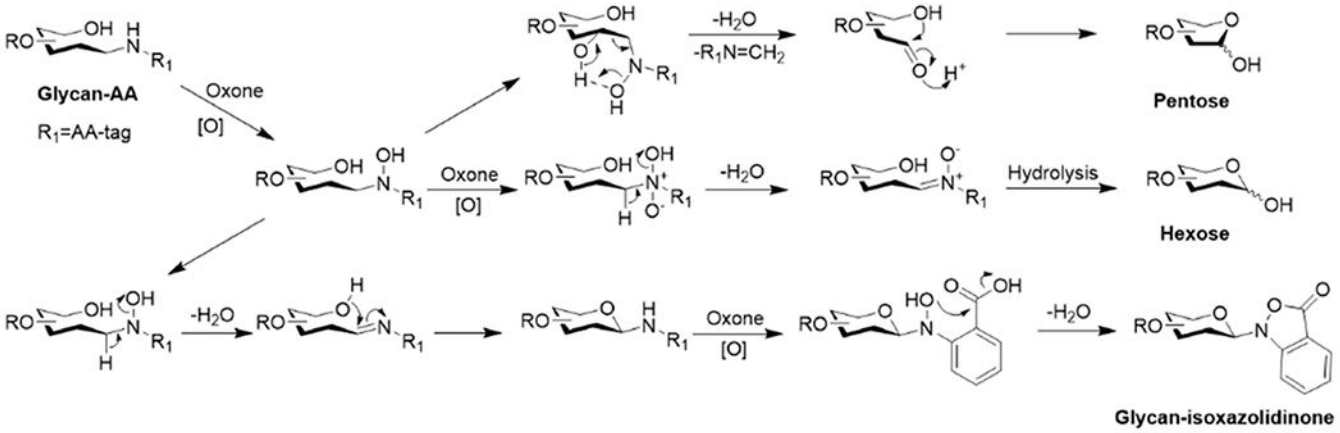


Figure 3.
Proposed mechanism of Oxone-catalyzed tag removal of glycan-AA conjugates by Oxone.

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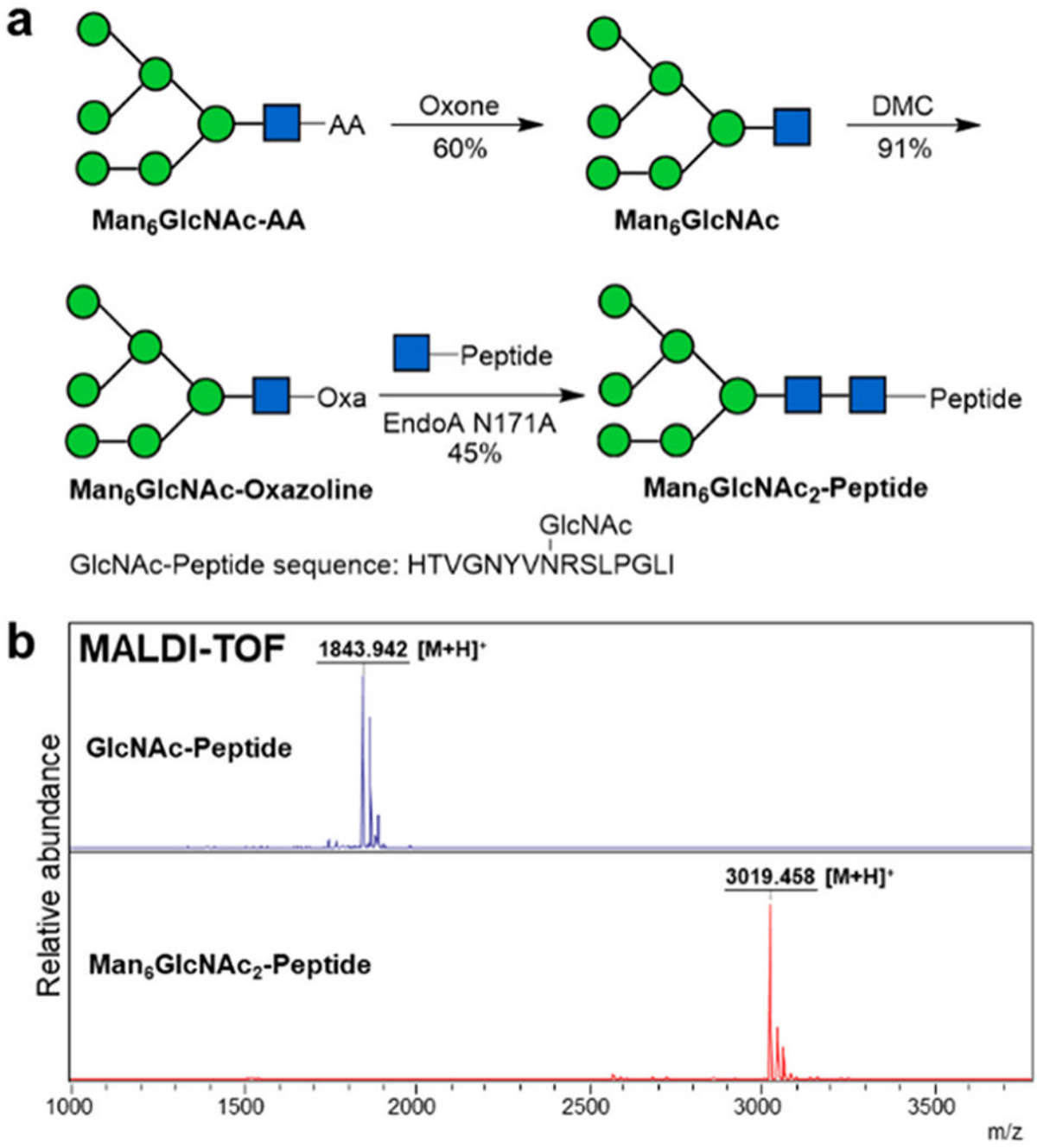


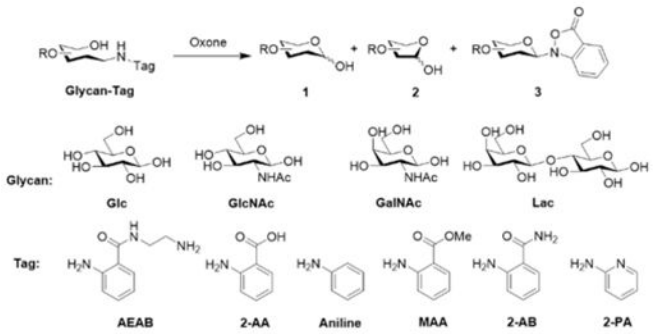
Figure 4.

(a) Reaction scheme for the synthesis of $Man_6GlcNAc_2$-peptide. (b) MALDI-TOF spectra of the peptide before and after enzymatic reaction.

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Table 1.

Oxone Treatment of Glycan Conjugates with Various Tags[a]



			yield[b] (%)		
entry	**glycan**	**tag**	**1**	**2**	**3**
1	Glc	AEAB	31	28	41
2	GlcNAc	AEAB	20	0	80
3	GalNAc	AEAB	74	0	26
4	Lac	AEAB	27	45	28
5	Glc	2-AA	63	17	20
6	GlcNAc	2-AA	75	0	25
7	GalNAc	2-AA	86	0	14
8	Lac	2-AA	66	22	12
9	Glc	aniline	48	52	0
10	GlcNAc	aniline	100	0	0
11	GalNAc	aniline	100	0	0
12	Lac	aniline	46	54	0
13	Glc	MAA	66	14	20
14	GlcNAc	MAA	54	0	46
15	GalNAc	MAA	89	0	11
16	Lac	MAA	56	30	14
17	Glc	2-AB	53	16	31
18	GlcNAc	2-AB	52	0	48
19	Glc	2-PA	0	0	0
20	GlcNAc	2-PA	0	0	0

[a]Reaction conditions: 1 mg of glycan tag treated with 1 mL of 1% Oxone in a 0.1% TFA solution.

[b]Yields were determined by ^{1}H NMR.